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Manufacture.Database

SQL Server Data Tools (SDK-style) database project that is the source of truth for the Manufacturing application's relational schema.

The project is built with `Microsoft.Build.Sql` and produces a `.dacpac` that can be published to SQL Server, Azure SQL, or a local SQL Server LocalDB instance.

Schemas

Schema	Responsibility
`core`	Login accounts and audit trail.
`hr`	Employees, salary advances, and monthly payroll.
`catalog`	Suppliers, inventory items, raw ingredients, recipes, and recipe lines.
`production`	Production batches, factory defects, and store/material requests.
`sales`	Customers, POS terminals, sales reps, orders, supermarket accounts, customer returns, and daily route sheets.

Schema	Responsibility
`logistics`	Drivers, vehicles, fuel logs, route rates, charter bookings, trip legs, maintenance, and weekly fleet expenses.
`finance`	Treasury policy settings.

Table map

core

- `Users` — application logins, roles, PIN/password hashes.
- `AuditLogs` — append-only change history.

hr

- `Employees` — staff master record with emergency and guarantor contacts.
- `SalaryAdvances` — staff loans and repayment balances.
- `PayrollRecords` — one row per employee per payroll month.

catalog

- `Suppliers`
- `InventoryItems`
- `Ingredients` — raw materials priced per unit.
- `Recipes` — sellable products and their baseline-yield costing header.
- `RecipeIngredients` — ingredient lines for one baseline batch.

production

- `ProductionBatches` — logged production runs with costing snapshots.
- `ProductionDefects` — factory rejects removed before loading.
- `StoreRequests`
- `StoreRequestItems`

sales

- `Customers`
- `PosTerminals`
- `SalesReps` — field roster, route, POS terminal, and vehicle assignment.
- `Orders`
- `OrderItems`
- `SupermarketAccounts`
- `SupermarketAccountPrices` — agreed per-product prices for an account.
- `CustomerReturns` — prior-sale items returned by customers.
- `DailyRepRouteSheets` — daily rep reconciliation header.
- `RouteSheetItems` — per-SKU loaded, sold, closing-stock, and damaged quantities.

logistics

- `Drivers`
- `Vehicles`
- `FuelLogs`
- `RouteRates`
- `CharterBookings`
- `TripLegs`
- `MaintenanceLogs`
- `WeeklyFleetExpenses`

finance

- `TreasuryPolicySettings` — single-row cash/digital tender policy.

Core relationships

```
Parse error on line 1:
erDiagram    Users
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Conventions

- Tables are plural and PascalCase; schemas are lowercase.
- Every table has an `Id` surrogate key (`INT IDENTITY`), except audit rows (`BIGINT`) and the single-row settings table.
- Business keys are protected by unique constraints or filtered unique indexes (for example `Users.Email`, `SalesReps.UserId`, `Ingredients.ItemCode`).
- Enum-like columns are `INT` with a named `CHECK` constraint so the numeric values match the C# enums in `Manufacture.Library/Models/Entities`.
- Money uses `DECIMAL(18,2)`; quantities use `DECIMAL(18,3)`; unit costs use `DECIMAL(18,4)`.
- Derived values that are safe to persist (`LineTotal`, `Variance`, `RemainingBalance`, `NetSalary`, and similar) are persisted computed columns.
- Child line tables cascade from their parent header; reference/master tables use `NO ACTION` to avoid accidental deletes and multiple cascade paths.
- `CreatedAt` / `UpdatedAt` default to `SYSUTCDATETIME()`; the application is responsible for refreshing `UpdatedAt` on updates.
- Aggregate roots that benefit from optimistic concurrency carry `ROWVERSION`.

Build

```
dotnet build Manufacture.Database/Manufacture.Database.sqlproj
```

The compiled model is written to `bin/Debug/Manufacture.Database.dacpac`.

Deploy to LocalDB

The repository includes a helper that builds, creates the database, publishes the schema, and applies the baseline seed rows:

```
pwsh scripts/deploy-database.ps1
```

Defaults:

- Server: `(localdb)\MSSQLLocalDB`
- Database: `ManufactureDb`

Override them when needed:

```
pwsh scripts/deploy-database.ps1 -Server "localhost" -Database "ManufactureDb_Dev"
```

`Microsoft.SqlPackage` and `dotnet-ef` are pinned as local tools in `dotnet-tools.json`; run `dotnet tool restore` after cloning.

Re-scaffold the EF Core models

After changing the schema, regenerate the DbContext and entities in the class library:

```
pwsh scripts/scaffold-database.ps1
```

Generated files live in `Manufacture.Library/Data`:

- `ManufactureDbContext.cs`
- `Scaffolded/*.cs`

The hand-written domain models and DTOs remain in `Manufacture.Library/Models`.